Viscan Voice Dental Assistant

Voice-first report retrieval and dictation drafting for dental imaging

Dr. Enosh A Paulson
github.com/DrEnosh/Vi-scan
Python · FastAPI · Groq (gpt-oss-120b, Whisper)

DESIGN PRINCIPLE

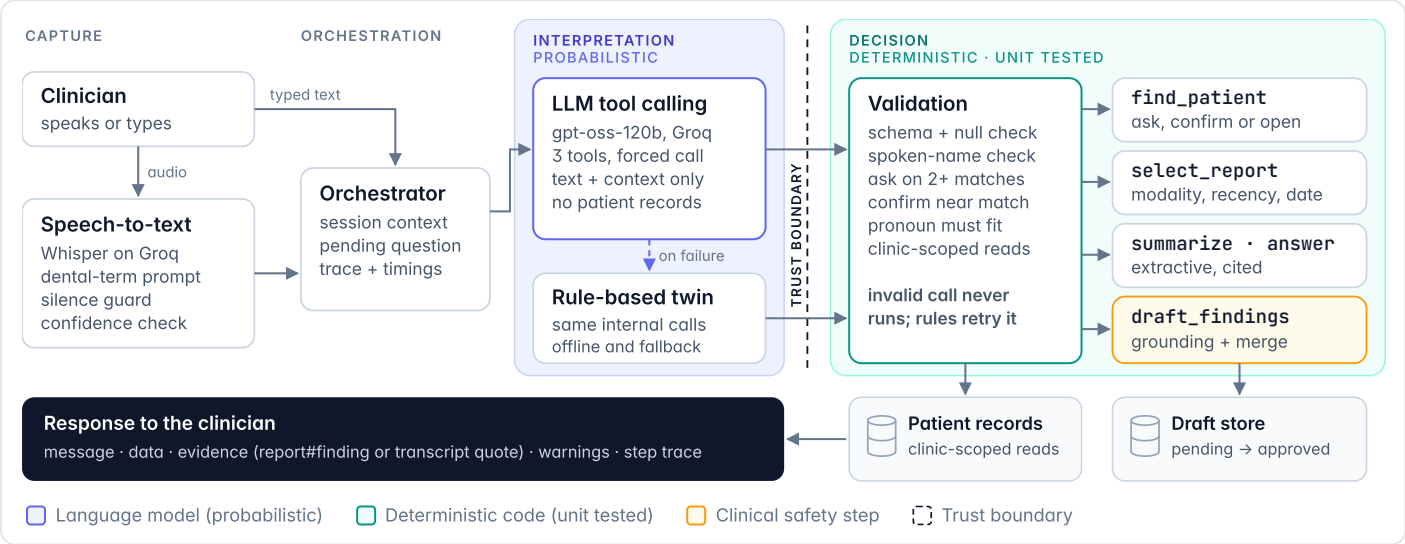
The model interprets. Code decides.

A model output is only a proposal. Patient identity, report selection and every draft finding are checked by tested, deterministic code before the clinician sees them. Nothing enters a report that was not said.

≈1.3 s	9 / 9	89 + 4	3	$0.0003
Spoken command, server side (live mean)	Correct tool calls in the live run	Automated tests + live API tests, all passing	Tools the model can call; everything else is code	Per spoken command at paid API rates

01 System architecture

Arrows show the request path; the dashed line is the trust boundary



02 Request flow

1 Voice to text
The browser records audio and posts it to `/api/assistant/voice`. Whisper transcribes it with a dental-term prompt; known silence hallucinations are rejected and low-confidence transcripts flagged. Typed text skips this step.

2 Interpretation
The utterance and a one-line context go to the model, which must call exactly one of three tools. It never sees patient records. On a timeout, rate limit or unusable reply, the rule-based twin makes the same call.

3 Validation and tools
Arguments are schema-checked and an invalid call never runs. Code restores a dropped spoken name, resolves the patient (ask, confirm or open), checks pronouns, enforces clinic scope and selects the report.

4 Report generation and evidence
Each dictated finding must quote the transcript and name the finding; tooth and negation are re-derived. A parser adds missed findings. Drafts stay pending until approved, and summaries cite `report_id#finding_id`.

03 Production considerations

CONCERN	IN THIS PROOF OF CONCEPT	NEXT STEP FOR PRODUCTION
Privacy	The model sees the utterance and a one-line context only; the audit log keeps IDs, not clinical text. Synthetic data.	Provider DPA or in-region self-hosting (DPDP Act), zero retention, encrypted drafts, PHI-free logs.
Authorization	Clinic-scoped reads; out-of-scope patients look non-existent; sessions bound to one clinician; drafts private.	OIDC tokens in place of the demo header, role-based sign-off, break-glass access logging.
Latency	Live: STT 0.35–0.48 s, model 0.59–1.61 s, code ≈1 ms. An 8 s model timeout falls back to rules at once.	Streaming STT; rules-first routing so the model handles dictation and unclear phrasing only.
Cost	Free on Groq's free tier, at about 860 input tokens per model call.	About $0.0003 per spoken command at paid rates; self-host Whisper and an open-weight model at volume.
Failure handling	Model error or HTTP 429: rule fallback. STT failure: clear 4xx. Input limits; no stack traces to clients.	Retry budgets, a circuit breaker, and alerts on rejection and clarification rates.